

Environment Shapes Emotional Cognitive Abilities More Than Genes

Featured

Genetics

Neuroscience

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Summary: Environmental factors may have a more significant impact on certain cognitive abilities than genetics. The study involved 57 pairs of identical and 48 pairs of fraternal twins from the Beijing Twin Study, focusing on metacognition and mentalizing—abilities tied to understanding and controlling one's cognitive processes and recognizing emotions in others, respectively.

Researchers revealed that twins, regardless of genetic similarity, showed similar cognitive traits when raised in similar educational and socio-economic environments. This challenges previous beliefs about the heritability of these cognitive skills, suggesting that the family environment plays a crucial role.

Key Facts:

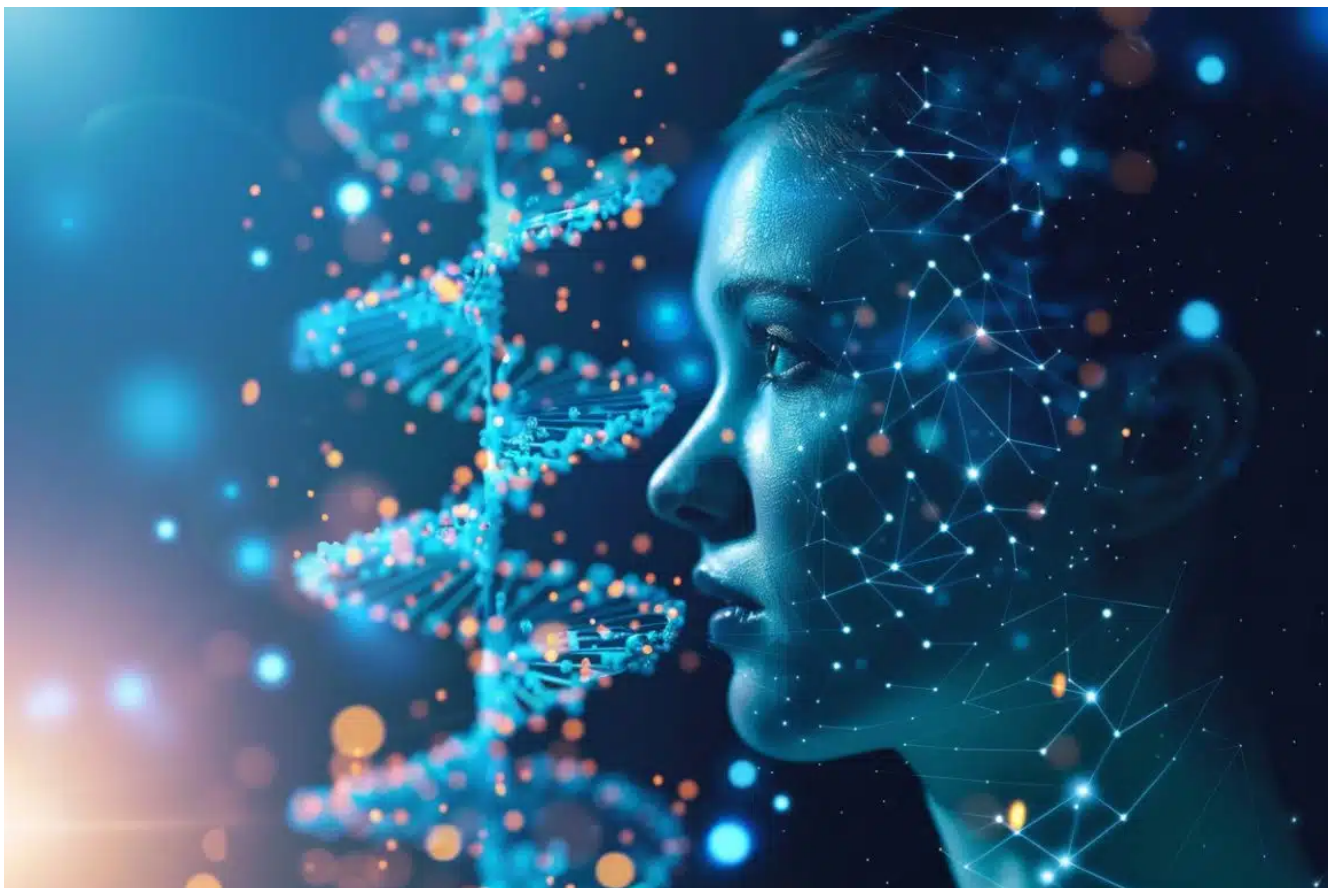
1. The study differentiated between general intelligence, which is significantly heritable, and other cognitive abilities like metacognition and mentalizing, which appear to be more influenced by environmental factors.
2. Participants performed tasks that measured their ability to assess motion direction and their confidence in decisions, along with their capacity to evaluate others' mental states.
3. Findings indicated that twins with parents who had higher education and income levels displayed similar metacognitive abilities, pointing to the influence of shared family environments over genetics.

Source: Cell Press

Twin studies have proven invaluable for teasing out the effects of both genetics and the environment on human biology.

In a study published April 2 in *Cell Reports*, researchers studied pairs of twins to look at how the interplay of genetics and environment affect cognitive processing—the way that people think.

They found that some cognitive abilities appear to be regulated more by environmental factors than by genetics.



Cognitive functions such as perception, attention, memory, language, and planning are considered to be the basis for general intelligence. Credit: Neuroscience News

“Past research has suggested that general intelligence—often referred to as intelligence quotient or IQ—has a heritability ranging from 50% to 80%,” says senior and corresponding author Xiaohong Wan of Beijing Normal University in China.

“Our study may be the first to demonstrate that a different kind of cognitive ability, known as metacognition and mentalizing, might be much more influenced by environment.”

Cognitive functions such as perception, attention, memory, language, and planning are considered to be the basis for general intelligence. These functions regulate the way that people organize and process new information. By contrast, metacognition looks at how well people understand and control their cognitive processes.

Metacognition is important for developing learning strategies and is believed to be a predictor of an individual’s successes in school and social achievements. Mentalizing describes the process of recognizing and understanding mental states like emotions and attitudes, both in ourselves and in other people.

For this research, the investigators recruited 57 pairs of adult monozygotic (identical) twins and 48 pairs of dizygotic (fraternal) twins from the Beijing Twin Study (BeTwiSt).

This is an ongoing, long-term study established in 2006 that includes extensive data like brain images and psychological surveys, as well as genetic information, on pairs of twins.

The twins were asked to perform tasks related to metacognition. These tasks consisted of watching a cluster of moving dots on a screen and making a perceptual judgement on the net direction of the dots.

They were also asked to rate their confidence about their decisions. To measure mentalizing, the participants were asked to evaluate a partner’s confidence in their decision-making abilities.

The investigators found that pairs of twins who had parents with higher levels of education and higher family incomes have similar results to each other, regardless of whether they were identical or fraternal. These observations suggested that familial environment was more likely to influence metacognitive abilities than genetics.

“Our findings were outside our expectations,” Wan says.

“Decades of extensive research utilizing the classical twin paradigm have consistently demonstrated the heritability of nearly all cognitive abilities so far investigated.

“Our findings emphasize that these shared family environmental factors, such as parental nurturing and the transmission of cultural values, likely play a significant role in shaping the mental state representations in metacognition and mentalizing.”

The researchers acknowledge that there are limitations to this research and that many more studies are needed. They plan to continue their research in this area, including using population studies to further investigate what kind of specific parental nurturing factors and sociocultural values affect individuals’ metacognitive and mentalizing abilities.

Funding: This research was supported by the Ministry of Science and Technology of the People’s Republic of China, the National Natural Science Foundation of China, the Interdisciplinary Innovation Team of the Chinese Academy of Sciences, and the BeTwiSt of Institute of Psychology, Chinese Academy of Sciences.

About this neuroscience and metacognition research news

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Original Research: Open access.

[“Distinct Genetic and Environmental Origins of Hierarchical Cognitive Abilities in Adult Humans”](#) by Xiaohong Wan et al. *Cell Reports*

Abstract

Distinct Genetic and Environmental Origins of Hierarchical Cognitive Abilities in Adult Humans

Highlights

- Human cognitive abilities can be categorized into a two-order hierarchy
- First-order cognitive abilities are primarily influenced by genetic factors
- Second-order cognitive abilities are more influenced by environmental factors
- Multiple family factors contribute to second-order cognitive abilities

Summary

Human cognitive abilities ranging from basic perceptions to complex social behaviors exhibit substantial variation in individual differences.

These cognitive functions can be categorized into a two-order hierarchy based on the levels of cognitive processes. Second-order cognition including metacognition and mentalizing monitors and regulates first-order cognitive processes.

These two-order hierarchical cognitive functions exhibit distinct abilities. However, it remains unclear whether individual differences in these cognitive abilities have distinct origins.

We employ the classical twin paradigm to compare the genetic and environmental contributions to the two-order cognitive abilities in the same tasks from the same population.

The results reveal that individual differences in first-order cognitive abilities were primarily influenced by genetic factors. Conversely, the second-order cognitive abilities have a stronger influence from shared environmental factors.

These findings suggest that the abilities of metacognition and mentalizing in adults are profoundly shaped by their environmental experiences and less determined by their biological nature.

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One Comment

Julia

April 26, 2024 at 8:37 am

Nature versus Nurture, stop trying to change science that has already been discussed and proven. No

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